

```
R Console
[Previously saved workspace restored]

> A <- c(18, 2, 20)
> B <- c(22, 28, 10)
> C <- c(20, 40, 40)
>
> cov(B, C)
[1] -20
>
> data <- data.frame(A, B, C)
> cov(data)
      A      B      C
A 97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
>
> cor(B, C)
[1] -0.1889822
>
> cor(data)
      A      B      C
A 1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
>

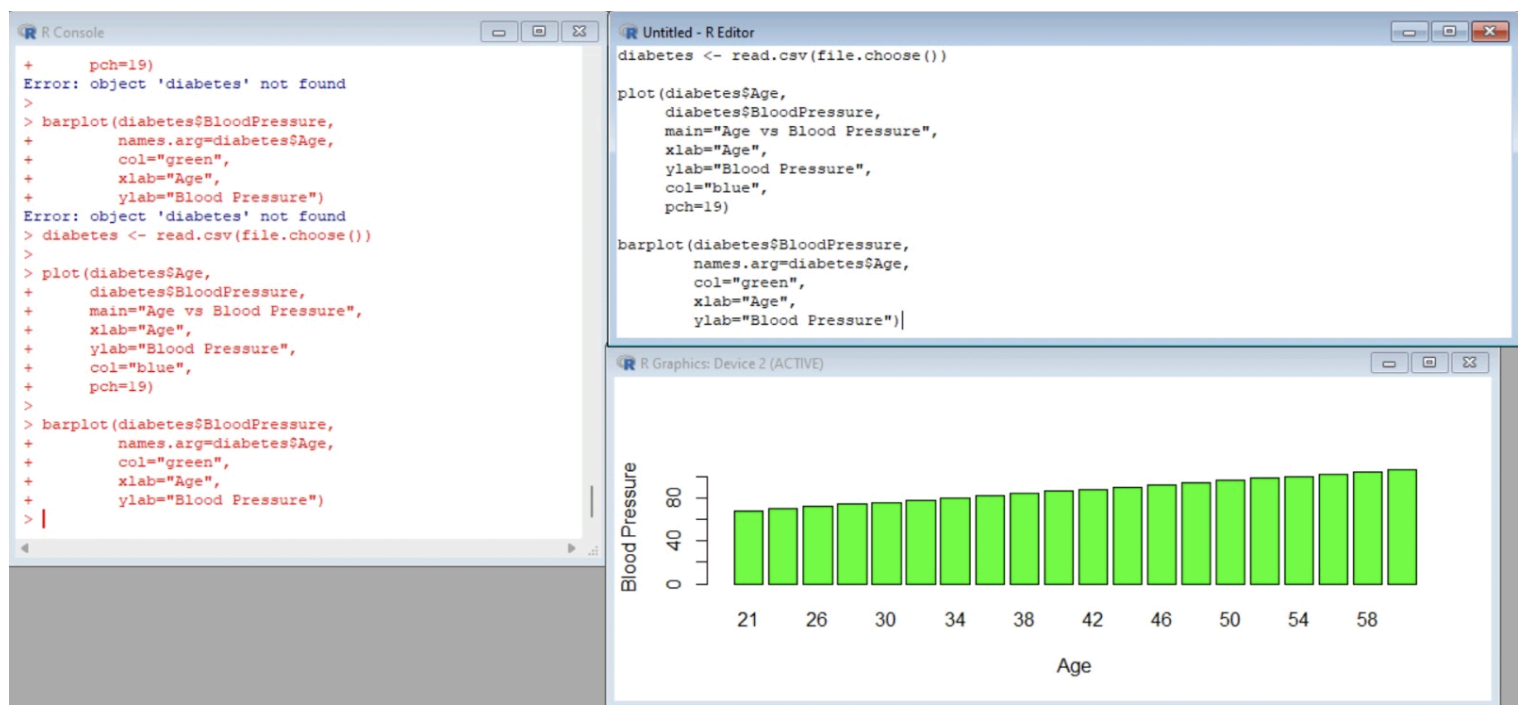
Untitled - R Editor
A <- c(18, 2, 20)
B <- c(22, 28, 10)
C <- c(20, 40, 40)

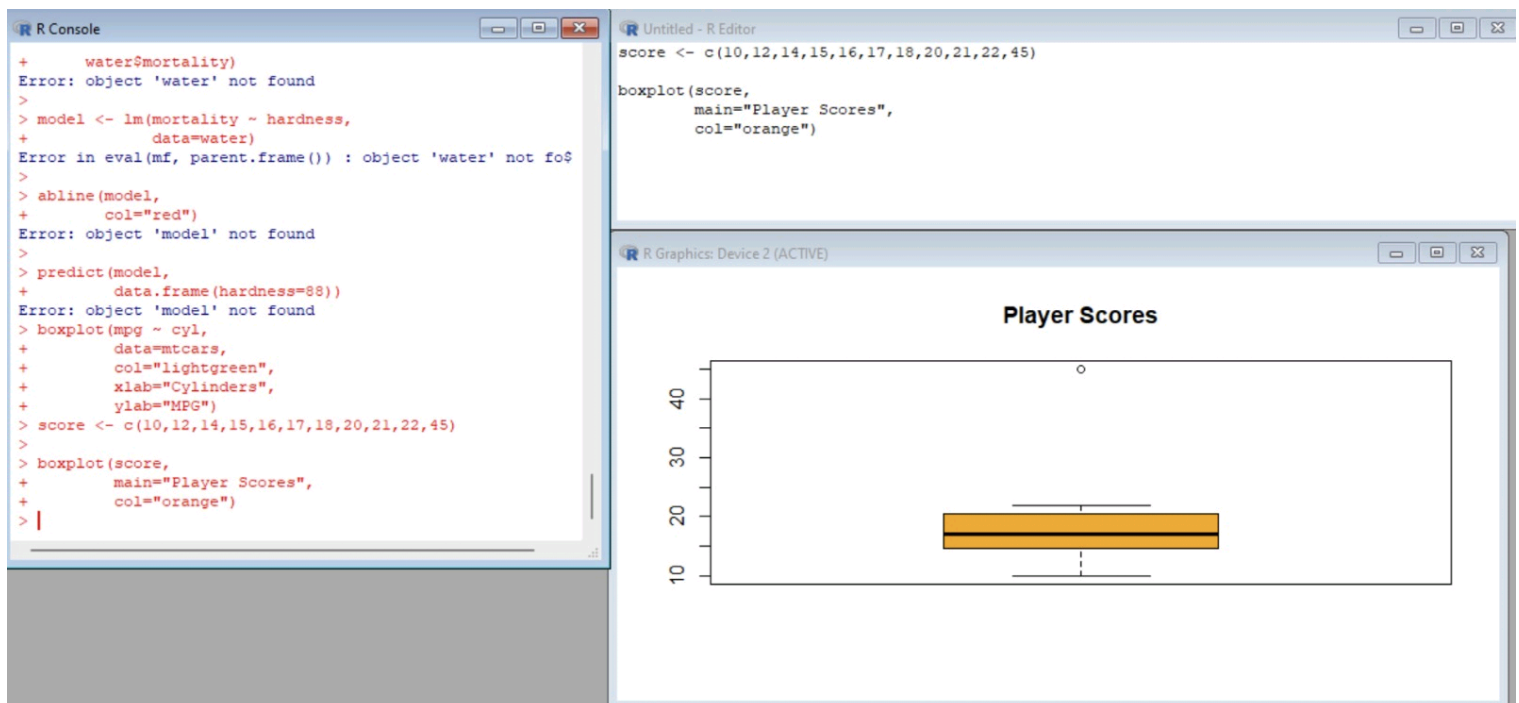
cov(B, C)

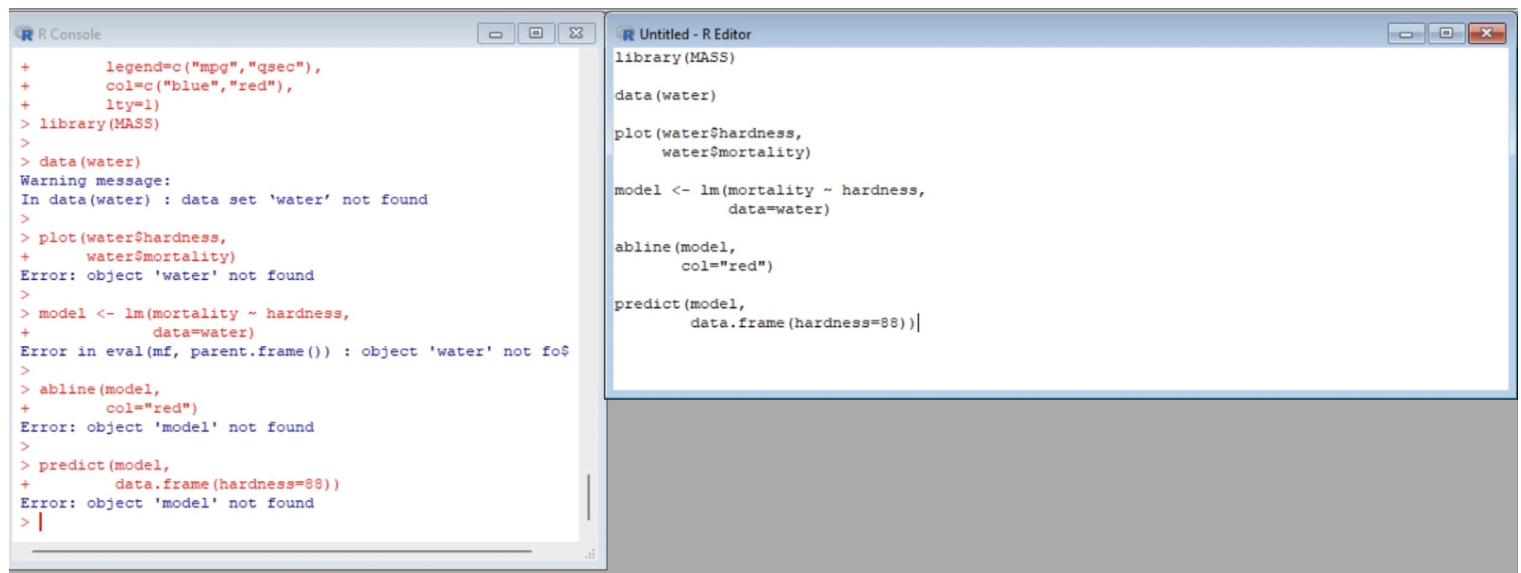
data <- data.frame(A, B, C)
cov(data)

cor(B, C)

cor(data)
```







The image shows two windows from the RStudio interface. The 'R Console' window on the left contains the following code and output:

```
+ legend=c("mpg","qsec"),
+ col=c("blue","red"),
+ lty=1)
> library(MASS)
>
> data(water)
Warning message:
In data(water) : data set 'water' not found
>
> plot(water$hardness,
+      water$mortality)
Error: object 'water' not found
>
> model <- lm(mortality ~ hardness,
+            data=water)
Error in eval(mf, parent.frame()) : object 'water' not found
>
> abline(model,
+        col="red")
Error: object 'model' not found
>
> predict(model,
+         data.frame(hardness=88))
Error: object 'model' not found
> |
```

The 'Untitled - R Editor' window on the right contains the following code:

```
library(MASS)

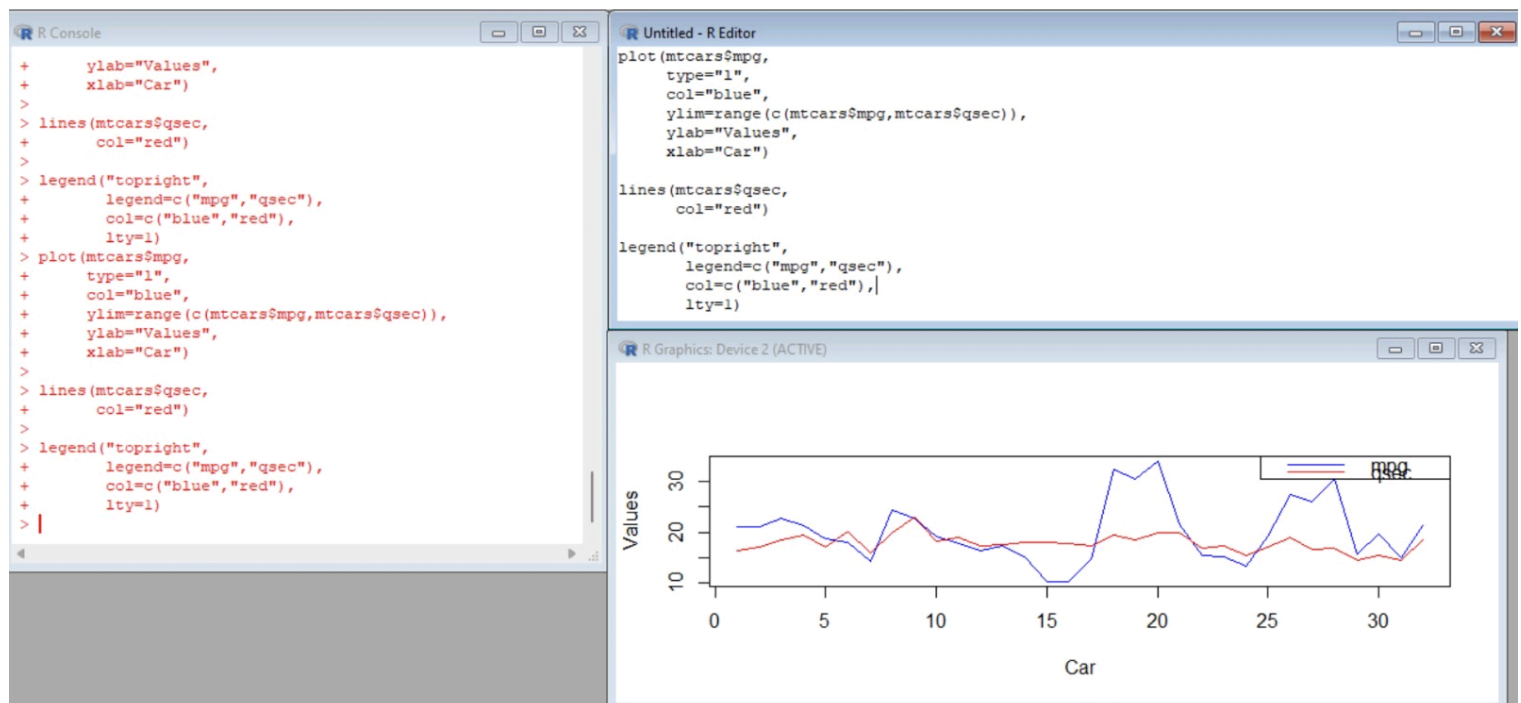
data(water)

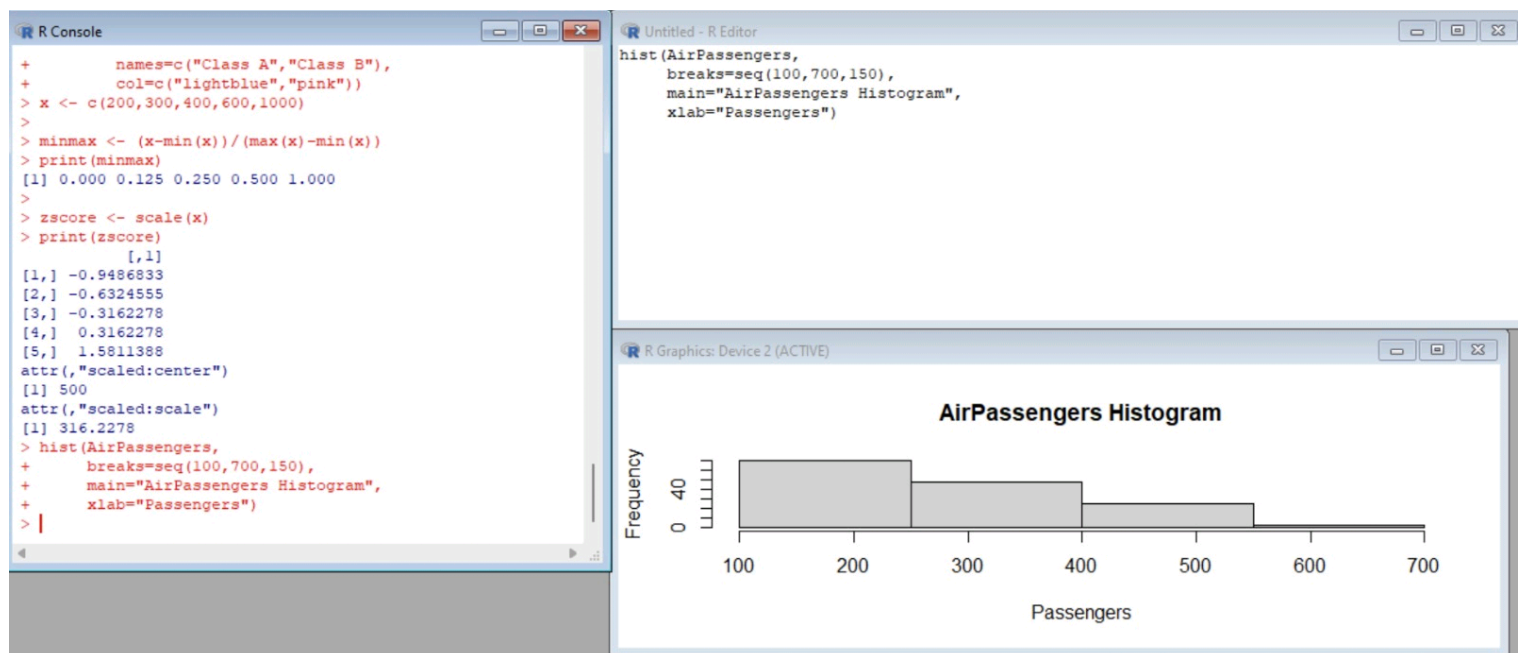
plot(water$hardness,
     water$mortality)

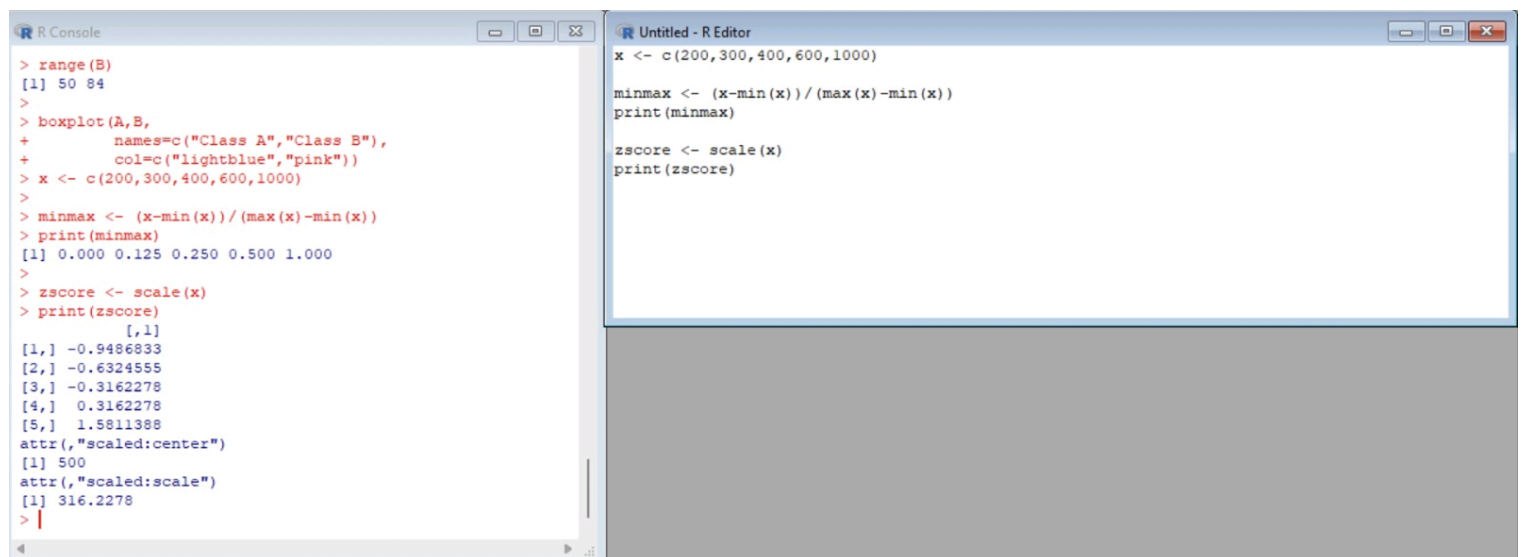
model <- lm(mortality ~ hardness,
           data=water)

abline(model,
       col="red")

predict(model,
       data.frame(hardness=88))|
```







The image shows a screenshot of an R environment with two windows: 'R Console' and 'Untitled - R Editor'.

R Console Output:

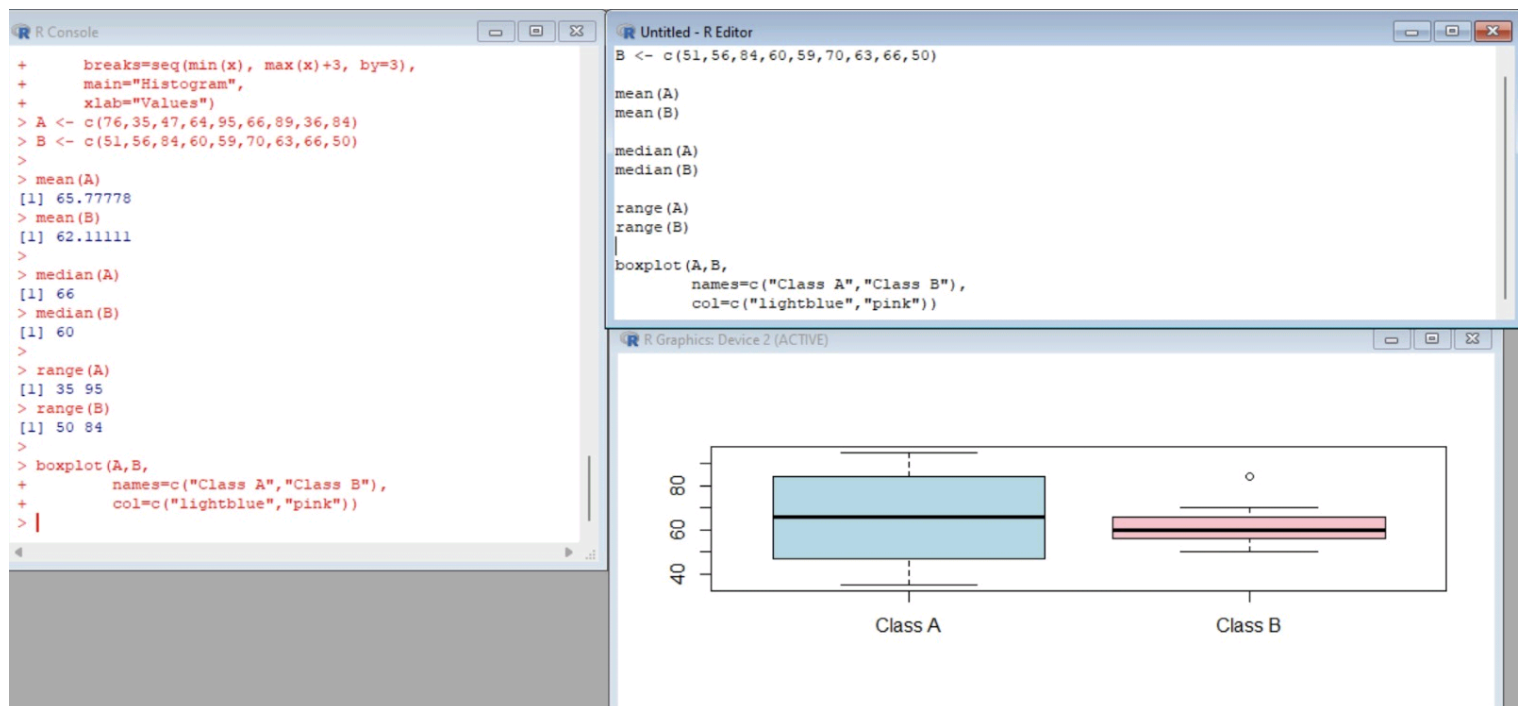
```
> range(B)
[1] 50 84
>
> boxplot(A,B,
+         names=c("Class A","Class B"),
+         col=c("lightblue","pink"))
> x <- c(200,300,400,600,1000)
>
> minmax <- (x-min(x))/(max(x)-min(x))
> print(minmax)
[1] 0.000 0.125 0.250 0.500 1.000
>
> zscore <- scale(x)
> print(zscore)
      [,1]
[1,] -0.9486833
[2,] -0.6324555
[3,] -0.3162278
[4,]  0.3162278
[5,]  1.5811388
attr(,"scaled:center")
[1] 500
attr(,"scaled:scale")
[1] 316.2278
> |
```

Untitled - R Editor Code:

```
x <- c(200,300,400,600,1000)

minmax <- (x-min(x))/(max(x)-min(x))
print(minmax)

zscore <- scale(x)
print(zscore)
```




```
$`11`  
[1] 20  
  
$`12`  
[1] 20.666667  
  
$`13`  
[1] 22.333333  
  
$`14`  
[1] 25  
  
$`15`  
[1] 27  
  
$`16`  
[1] 30
```

```
> hist(x,  
+       breaks=seq(min(x), max(x)+3, by=3),  
+       main="Histogram",  
+       xlab="Values")  
>
```

```
x <- c(1,1,5,5,5,5,5,8,8,10,10,10,10,12,14,14,14,15,15,15,15,15,15,18,18,18,18,18,20,20,20,20,.)  
bins <- split(x, ceiling(seq_along(x)/3))  
print(bins)  
  
bin_mean <- lapply(bins, mean)  
print(bin_mean)  
  
hist(x,  
      breaks=seq(min(x), max(x)+3, by=3),  
      main="Histogram",  
      xlab="Values")
```

